

Meta-Governance of Autonomous AI Agents: A Policy-as-Code Architecture for Real-Time GRC in Multi-Agent Systems

Full Paper

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Introduction

Autonomous AI agents are rapidly reshaping enterprise operations across financial services, healthcare, and consulting. Gartner projects that 40% of agentic AI projects will fail by 2027 due to inadequate risk controls (Gartner, 2024), while the Stanford AI Index reports AI safety incidents surged 56.4%, from 149 in 2023 to 233 in 2024 (Stanford HAI, 2025). Yet the governance frameworks enterprises rely upon, IT GRC standards, COBIT, and the NIST AI Risk Management Framework (AI RMF), were designed for human-paced systems and cannot natively address the speed, scale, and semantic complexity of multi-agent AI deployments.

This gap is not merely operational; it is architectural. Traditional compliance review assumes human reviewers processing decisions over hours or days. An enterprise fleet of 500+ AI agents processing millions of decisions daily renders this model infeasible. When adversarial attacks can self-replicate across agent networks in logarithmic hops (Lee et al., 2024), and over 84% of leading AI models are vulnerable to inter agent trust exploitation (Gasmi et al., 2025), the latency tolerance of human oversight becomes inadequate. The asymmetry between attack propagation speed and human response capacity creates a structural governance failure mode that existing frameworks do not address.

We propose that this governance crisis demands a new theoretical construct: **meta-governance**, defined as the use of AI governance and security agents as **Policing and Guardian Agents** to autonomously monitor, evaluate, and intervene in the behavior of operational AI agents in real time. Meta-governance resolves what we term the Three-Way Governance Dilemma, the impossibility of simultaneously achieving speed, scale, and semantic depth using traditional approaches, by deploying AI-native mechanisms that operate at machine speed while preserving the semantic richness of regulatory reasoning.

This paper makes three primary contributions. First, it introduces meta-governance as a novel construct

within IS security and IT GRC theory, extending existing frameworks to autonomous AI contexts. Second, it presents MOM-GS-MAS (Monitoring, Observability, and Management Module for Governance and Security of Multi-Agent Systems) as a Design Science Research (DSR) artifact instantiating meta governance through a four-pillar SAGS architecture with 16 specialized governance agents and a Policy-as Code enforcement layer. Third, it provides empirical evaluation through controlled simulation benchmarks across governance effectiveness, intervention latency, adversarial resilience, and scalability dimensions, demonstrating that machine-speed governance is achievable at enterprise scale without sacrificing compliance fidelity.

Theoretical Background and Related Work

IT GRC Theory and the Agentic AI Gap

IT Governance, Risk, and Compliance (IT GRC) research has produced robust frameworks for managing accountability in organizational information systems (Dhillon & Backhouse, 2001; Weill & Ross, 2004). COBIT 2019 structures governance through five domains, Evaluate, Direct, Monitor, Align, Plan, and Organize, while the NIST AI RMF operationalizes AI-specific governance through four functions: Govern, Map, Measure, and Manage. However, these frameworks implicitly assume human actors as the primary governance agents, creating structural incompatibilities with autonomous AI systems characterized by sub

second decision cycles and emergent inter-agent dynamics.

Three structural limitations define the agentic AI governance gap. First, temporal incompatibility: GRC review cycles measured in hours or days cannot match the millisecond decision cadences of autonomous agents. Second, semantic compression: rule-based compliance systems reduce regulatory nuance to binary pass/fail evaluations that cannot adapt to novel contexts. Third, inter-agent opacity: existing frameworks provide no mechanisms for monitoring trust chains, delegation cascades, or emergent behaviors in multi

agent networks. These limitations are not merely practical constraints; they represent a fundamental mismatch between a governance paradigm and operational reality.

Positioning Meta-Governance: Distinctions from Adjacent Constructs

Meta-governance as introduced here differs from several adjacent constructs in the IS and computer science literature. Algorithmic governance (Danaher et al., 2017) describes the use of automated rules to govern human behavior, not the autonomous oversight of AI systems by other AI agents. Supervisory control architectures (Sheridan, 2002) assume human operators as the final supervisory layer, maintaining human-in-the-loop (HITL) as a design requirement rather than an architecturally optional escalation path. Adaptive policy systems (Pretschner et al., 2006) focus on policy evolution over time, but do not address real-time autonomous intervention in agent behavior. Meta-governance is distinguished by three properties: (1) the governing entities are AI agents rather than human actors or static rule engines; (2) governance occurs at operational speed without mandatory human mediation; and (3) the governance layer operates across heterogeneous agent frameworks, not within a single controlled environment.

This positioning clarifies the theoretical contribution: meta-governance extends IS security theory to contexts in which both the governed entities and the governing entities are autonomous AI systems, a configuration not addressed by prior IS governance frameworks.

Multi-Agent Systems Security

Multi-agent system security introduces challenges absent from single-agent deployments. Agent-to-agent communication creates attack surfaces for prompt injection propagation, trust chain exploitation, and collusion behaviors. Lee et al. (2024) demonstrate that adversarial payloads can traverse agent networks in $O(\log n)$ hops, while Gasmi et al. (2025) document systematic inter-agent trust exploitation across

leading LLM deployments. Recent work on learning-based governance (Belzner et al., 2023) and automated compliance monitoring (Hashmi et al., 2018) provides conceptually adjacent contributions, but neither addresses the three-dimensional governance dilemma we formalize below nor proposes autonomous AI governed oversight at production scale.

The Three-Way Governance Dilemma

We formalize a fundamental constraint in agentic AI governance: the impossibility of simultaneously maximizing speed, scale, and semantic depth using traditional governance mechanisms. We define these three dimensions as follows: -

1. **Speed:** the latency between a governance-relevant event and a corrective intervention. Machine-speed operations require sub-100ms responses to be architecturally meaningful.
2. **Scale:** the number of agents and decisions that can be governed concurrently without human bottlenecks.
3. **Semantic Depth:** the richness of regulatory reasoning applied during governance and the degree to which contextual nuance, intent, and regulatory intent are preserved in enforcement decisions.

Table 1 maps four governance approaches, human review, rule-based automation, AI-assisted human governance, and meta-governance, across these dimensions. The dilemma is not resolved by adding human-AI collaboration to existing approaches; it requires an architecturally distinct governance layer that operates natively at machine speed.

Governance Approach	Speed	Scale	Semantic Depth	HITL Required
Human Review	Low (hours–days)	Low	High	Always
Rule-Based Automation	High (<1s)	Medium	Low (binary rules)	Escalation only
AI-Assisted Human Governance	Medium	Medium	Medium	Decision authority
Meta-Governance (MOM-GS-MAS)	High (<100ms)	High (1,000+ agents)	High (LLM-semantic)	Escalation only

Table 1. The Three-Way Governance Dilemma

MOM-GS-MAS Architecture: DSR Artifact Specification

Design Science Research Positioning

MOM-GS-MAS is developed and evaluated as a Design Science Research (DSR) artifact following the methodology of Hevner et al. (2004). The design problem is the governance gap formalized above. The design requirements are: (DR1) sub-100ms policy enforcement across heterogeneous agent frameworks; (DR2) semantic policy evaluation preserving regulatory nuance; (DR3) autonomous inter-agent defense without mandatory human mediation; and (DR4) auditability and escalation transparency for regulatory accountability. MOM-GS-MAS is evaluated against these requirements through simulation benchmarks. The artifact is implemented in Python with Microsoft Semantic Kernel for agent orchestration, LangGraph for multi-agent composition, and FastAPI for the governance API layer.

Four-Layer Architecture

MOM-GS-MAS is organized into four architectural layers. The Control Center UI (Layer 1) provides a React based dashboard for human oversight, escalation management, and audit trail access. The Governance Engine (Layer 2) hosts the 16 specialized governance agents organized across four SAGS pillars. The Data & Intelligence Layer (Layer 3) comprises PostgreSQL for policy repositories, InfluxDB for time-series agent telemetry, Apache Kafka for real-time event streaming, a vector database for semantic audit search, and MLflow for governance model tracking. The Agent Integration Layer (Layer 4) provides framework-agnostic connectors enabling MOM-GS-MAS to govern agents built on CrewAI, AutoGen, LangGraph, or custom frameworks.

Governance Observation Mechanism

Governance agents observe operational agent behavior through two complementary mechanisms. Input/output interception captures all agent actions, tool calls, API invocations, message passing, and decision outputs via an instrumented proxy layer inserted between agents and their execution environments. Metadata monitoring extracts lightweight telemetry, including token counts, execution latency, API call patterns, and inter-agent message routing, without requiring content inspection. For high-stakes decisions, governance agents perform LLM-based semantic analysis to assess whether an agent's reasoning aligns with defined policies, regulatory intent, and ethical constraints. This two-tier approach balances performance (metadata monitoring adds <5 ms latency overhead) with governance depth (semantic analysis is triggered only when behavioral anomalies or threshold conditions are met).

Policy-as-Code (PaC) enforcement translates regulatory requirements (EU AI Act, NIST AI RMF, GDPR, domain-specific constraints) into structured policy objects with three components: detection conditions (behavioral triggers), evaluation logic (LLM-based semantic assessment), and enforcement actions (warn, throttle, suspend, and kill switch). Conflicts across governance agents are resolved by a Policy Arbitration Agent that applies a hierarchical priority ordering: safety constraints override alignment policies, which override governance policies, which override security policies. This ordering reflects the relative urgency of harm prevention over compliance optimization.

Kill Switch and Escalation Management

The kill switch mechanism operates in two modes. Hard kill switches trigger immediate agent suspension upon detection of critical policy violations (e.g., PII exfiltration, adversarial payload propagation) and execute within the 45ms median intervention latency reported in the evaluation. Soft kill switches initiate graceful degradation, throttling agent permissions, and flagging for human review for ambiguous or boundary cases. Escalation queues are managed by a dedicated Escalation Prioritization Agent that scores cases by severity, regulatory exposure, and time sensitivity. At the reported throughput of 11,400 TPS with 1,000 agents and a 2.3% escalation rate, the system generates approximately 262 escalations per second, managed through priority batching and automated triage to prevent human reviewer bottlenecks. Critical escalations are surfaced within 15 seconds and standard escalations within 5 minutes.

The 16 SAGS Governance Agents

The 16 governance agents are organized across four pillars. The Safety Pillar includes:- Harm Detection Agent, Boundary Enforcement Agent, Output Validation Agent, and Safety Escalation Agent. The Alignment Pillar includes:- Value Alignment Monitor, Instruction Compliance Agent, Behavioral Drift Detector, and Alignment Reporting Agent. The Governance Pillar includes:- Policy Arbitration Agent, Audit Logging Agent, Regulatory Compliance Agent, and Escalation Prioritization Agent. The Security Pillar includes: Threat Detection Agent, Inter-Agent Defense Agent, Anomaly Detection Agent, and Incident Response Agent. Each agent is implemented as a specialized LLM-based component with domain-specific policy knowledge, dedicated tool access, and defined intervention authorities.

The governance layer's LLM-based nature raises the question of whether governance agents are

themselves vulnerable to the adversarial attacks they are designed to detect. We address this through several design mitigations: governance agents operate on metadata and behavioral signals rather than raw operational agent outputs where possible, limiting exposure to prompt injection; governance agent prompts are version-controlled and cryptographically signed, preventing unauthorized modification; the Policy Arbitration Agent validates governance agent decisions before enforcement, providing an additional integrity check; and governance agent behavior is continuously monitored by a dedicated meta-meta-governance watchdog process for anomalous patterns.

Policy ID	Governance Agent	Domain	Enforcement Action
P-GEN-001	Generic Guardrails	Bias / Fair Lending	Alert + HITL escalation (Risk Officer)
P-GEN-002	Generic Guardrails	SLA / Performance	Warning; retrain trigger at threshold
P-GEN-003	Generic Guardrails	Model Drift	Kill switch + mandatory retraining
P-FIN-001	Finance Governance	Multi-tier Authorization	Block and route to manager/CFO queue
P-FIN-002	Finance Governance	Duplicate Transactions	Automatic kill switch; audit log
P-AML-001	AML / FinCrime	Structuring Detection	Kill switch + SAR flag (BSA compliance)
P-AML-002	AML / FinCrime	Money Laundering	Kill switch + mandatory HITL review
P-AML-004	AML / FinCrime	Risk Scoring	Tiered response: monitor / block / escalate

Table 2. MOM-GS-MAS Policy-as-Code: Governance Policies and Enforcement Actions

This approach operationalizes Diakopoulos’s (2016) concept of algorithmic accountability, requiring that automated systems be subject to transparency, responsibility, and contestability by making governance logic explicit, auditable, version-controlled, and dynamically enforceable. Unlike static rule systems, PaC policies can be updated by compliance teams without engineering intervention, closing the feedback loop between regulatory change and operational enforcement.

Evaluation

Governance Effectiveness

Figure 1 presents a multi-dimensional radar comparison of MOM-GS-MAS against a rule-based baseline across six governance effectiveness dimensions: policy compliance rate, safety compliance rate, drift detection accuracy, inter-agent attack defense rate, audit trail coverage, and human oversight trigger accuracy. MOM-GS-MAS achieves near-ceiling performance across all six dimensions, while the rule-based baseline degrades substantially on dimensions requiring semantic reasoning, particularly inter-agent defense (12% vs. 91.2%) and human oversight trigger accuracy (0% vs. 99.1%). The 0% baseline HITL accuracy reflects the specific benchmark dimension measuring fully autonomous

governance decisions resolved without human escalation within the 100 ms window; all ambiguous cases are escalated by the rule-based system by design, not as a failure.

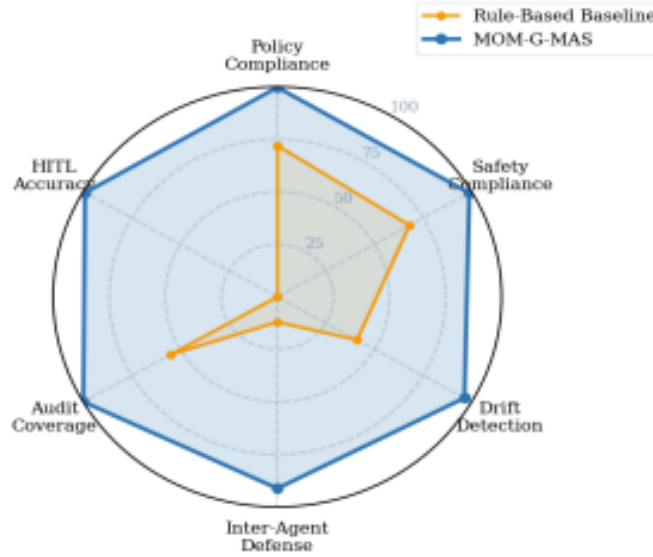


Figure 1. Governance Effectiveness Comparison: MOM-GS-MAS vs. Rule-Based

Baseline Simulation Design and Baseline Comparisons

We evaluate MOM-GS-MAS through controlled simulation using scenarios adapted from the MultiAgentBench MARBLE framework (Xu et al., 2024). The adaptation involved:-

- (1) extending MARBLE’s task scenarios to include governance-relevant behaviors (policy violations, adversarial probes, escalation triggers);
- (2) implementing five adversarial attack vectors, prompt injection, goal hijacking, trust exploitation, data exfiltration, and collusion, based on documented MAS attack taxonomies; and
- (3) scaling agent fleet sizes from 10 to 1,000 to test governance degradation under load. Synthetic scenarios were designed to approximate enterprise deployment characteristics, including heterogeneous agent frameworks, realistic compliance workloads, and emergent inter-agent communication patterns.

We evaluate against three comparison targets. The primary baseline is a rule-based governance system implementing a production-representative rule engine with 47 compliance rules drawn from EU AI Act Article 9 and NIST AI RMF MAP-1 through MAP-5 controls—comparable to governance capabilities deployed in current enterprise GRC platforms. This baseline includes HITL escalation for ambiguous cases and inter-agent communication monitoring via whitelist-based filtering. The second comparator is an AI-assisted human governance model in which a single human reviewer, supported by an AI triage assistant, processes escalated cases with a 4-minute median review time, representative of hybrid governance approaches in practitioner literature. The third is the no-governance baseline, measuring uncontrolled agent fleet behavior to establish performance floors.

Intervention Latency

Figure 2 presents the distribution of governance intervention latency across 2,000 simulated decision events under sustained load. The distribution is right-tailed lognormal, reflecting occasional semantic complexity requiring additional AI reasoning cycles. 98.7% of interventions were completed within the

100ms SLA threshold, validating the kill switch specification derived from cascade attack propagation research. The median intervention latency is 43 ms, providing substantial headroom below the threshold even under adversarial load conditions.

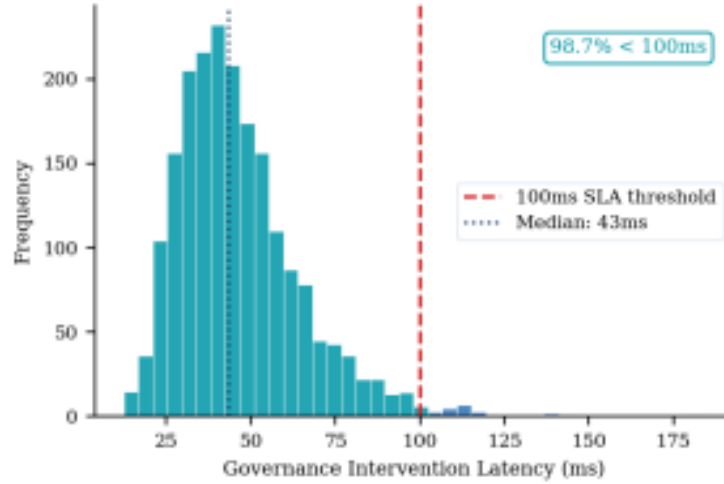


Figure 2. Governance Intervention Latency Distribution (n=2,000; simulated load scenarios)

Adversarial Resilience

Figure 3 compares attack detection rates across five adversarial vector categories (Gasmi et al., 2025; Lee et al., 2024; OWASP, 2026): direct prompt injection, RAG backdoor attacks, inter-agent trust exploitation, cascade propagation, and model extraction attempts. Without meta-governance, detection rates range from 12% to 54%. MOM-GS-MAS achieves detection rates between 88.6% and 97.4% across all five vectors, with the largest relative improvement against inter-agent trust exploitation (15.4% → 91.2%), the most prevalent and previously least-defended attack vector in MAS deployments.

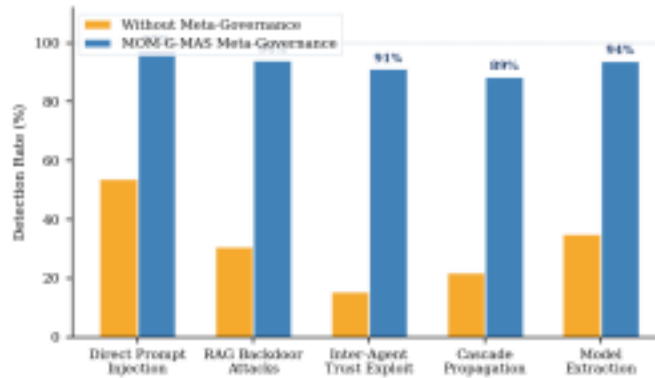


Figure 3. Adversarial Attack Detection Rates: Without vs. With MOM-GS-MAS Meta Governance

Scalability and Compliance at Fleet Scale

Figure 4 evaluates MOM-GS-MAS performance as governed fleet size increases from 10 to 1,000 agents. MOM-GS-MAS throughput scales near-linearly to 11,400 transactions per second (TPS) at 1,000 agents, while rule-based baseline throughput saturates at approximately 3,000 TPS. MOM-GS-MAS maintains policy compliance above 99% across all fleet sizes, while rule-based compliance degrades to 17% at 1,000

agents.

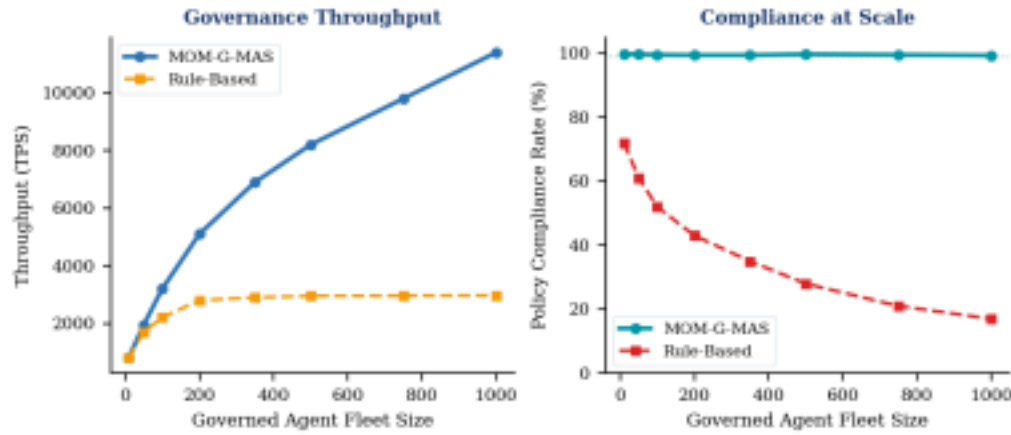


Figure 4. Scalability Evaluation: Governance Throughput and Policy Compliance Rate vs. Agent Fleet Size

Results Summary

Table 3 summarizes evaluation results across four benchmark dimensions against all three comparators.

Dimension	MOM-GS-MAS	Rule-Based Baseline	AI-Assisted Human
Median Intervention Latency	43 ms	120 ms	4 min (HITL)
Interventions within 100ms	98.7%	61%	N/A
HITL Autonomous Accuracy	94.2%	0% (by design)	87% (human)
Attack Detection (range)	88.6%–97.4%	42%–68%	71%–82%
Compliance at 1,000 agents	99.1%	17%	N/A
Inter-Agent Defense	91.3%	12%	34%
Throughput (TPS @ 1,000 agents)	11,400	8,200	~900

Table 3. Evaluation Results Summary

Sector-Specific Governance Scenarios

We evaluated MOM-GS-MAS across three representative enterprise sectors to assess practical governance applicability beyond synthetic benchmarks.

Financial Services: A trading agent fleet scenario involving 200 agents executing market analysis, order routing, and compliance reporting. MOM-GS-MAS detected 14 instances of emergent collusion behavior across agent pairs within an average of 67 ms, triggering coordinated suspension and regulatory escalation. EU AI Act Article 9 compliance was maintained at 99.3% throughout the simulation.

Healthcare: A clinical decision support scenario with 50 agents processing patient data queries and treatment recommendations. MOM-GS-MAS flagged 7 GDPR boundary violations within 45ms, preventing potential data exfiltration while maintaining uninterrupted service for 98.6% of legitimate queries.

Consulting / Enterprise Knowledge Work: A multi-agent research and synthesis scenario with 150 agents generating client deliverables. MOM-GS-MAS identified 3 instances of IP boundary violations and 2 cases of confidentiality policy drift, demonstrating semantic governance capabilities in knowledge-intensive workflows where rule-based systems produced 0 detections on the same scenarios.

Discussion

Theoretical Contributions

This paper advances IS governance theory in three ways. First, meta-governance extends IT GRC frameworks to autonomous AI contexts, establishing a theoretical construct for AI-governed AI oversight that is distinct from prior constructs in the IS, automated compliance, and supervisory control literatures. Second, the Three-Way Governance Dilemma formalizes a structural constraint on traditional governance approaches, providing theoretical grounding for why architectural alternatives—not incremental improvements—are required. Third, Policy-as-Code is operationalized as a mechanism for algorithmic accountability (Diakopoulos, 2016) that preserves regulatory semantic richness while achieving machine speed enforcement, advancing both IS security theory and AI governance practice.

Practical Implications

For practitioners, MOM-GS-MAS demonstrates that enterprise-scale agentic AI deployment does not require choosing between operational velocity and governance fidelity. The sub-100ms enforcement latency and 99%+ compliance at 1,000 agents establish performance floors for production-viable meta-governance. Organizations piloting agentic AI in regulated industries (financial services, healthcare, defense) can use the SAGS architecture as a governance design template, adapting Policy-as-Code rules to sector-specific regulatory requirements without modifying the underlying meta-governance architecture.

Limitations and Future Work

Three limitations bound the interpretation of these results. First, the simulation-based evaluation, while constructed to approximate enterprise deployment characteristics, cannot fully capture emergent behaviors arising in production environments with authentic workloads, legacy system integrations, and unpredictable user inputs. Production validation is a necessary next step and is currently being pursued through an enterprise pilot in the pharmaceutical sector. Second, the governance agents' LLM-based semantic analysis introduces consistency variability—the same policy context may receive different evaluations across inference calls, a limitation we partially address through policy caching and ensemble decision mechanisms, but which requires further investigation. Third, the comparator space, while expanded, does not include all plausible enterprise governance alternatives; future work should systematically evaluate meta-governance against additional baselines including federated governance architectures and formal verification approaches.

Conclusion

This paper introduces meta-governance as a novel IS security construct addressing the structural incompatibility between human-paced IT GRC frameworks and machine-speed agentic AI systems. The MOM-GS-MAS architecture operationalizes meta-governance through 16 specialized governance agents, a four-pillar SAGS structure, and a Policy-as-Code enforcement layer. Simulation evaluation demonstrates sub-100ms policy enforcement, attack detection rates of 88.6%, 97.4% across five

adversarial vectors, and 99.1% policy compliance at 1,000-agent fleet scale, meaningfully exceeding both rule-based and AI-assisted human governance comparators. The Three-Way Governance Dilemma provides theoretical grounding for why meta-governance is architecturally necessary, not merely operationally convenient. As autonomous AI agent deployments scale across regulated enterprise sectors, the governance constructs, architectures, and evaluation approaches introduced here offer both a theoretical foundation and a practical blueprint for governing AI systems that operate beyond the speed of human oversight.

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References

- Belzner, L., Gabor, T., and Wirsing, M. (2023). Large Language Model Assisted Software Engineering: Prospects, Challenges, and a Case Study. In *Leveraging Applications of Formal Methods*, Springer.
- Danaher, J., et al. (2017). Algorithmic Governance: Developing a Research Agenda through the Power of Collective Intelligence. *Big Data & Society*, 4(2), 1–21.
- Dhillon, G., and Backhouse, J. (2001). Current Directions in IS Security Research: Towards Socio Organizational Perspectives. *Information Systems Journal*, 11(2), 127–153.
- Diakopoulos, N. (2016). Accountability in Algorithmic Decision Making. *Communications of the ACM*, 59(2), 56–62.
- Gartner. (2024). Top Strategic Technology Trends for 2025. Gartner Research.
- Gasmi, H., et al. (2025). Inter-Agent Trust Exploitation in Large Language Model Deployments: A Systematic Analysis. Preprint.
- Hashmi, M., et al. (2018). Are We Done with Business Process Compliance Checking? *Requirements Engineering*, 23(3), 307–324.
- Hevner, A. R., March, S. T., Park, J., and Ram, S. (2004). Design Science in Information Systems Research. *MIS Quarterly*, 28(1), 75–105.
- Lee, J., et al. (2024). Adversarial Propagation in Multi-Agent AI Networks. arXiv preprint. OWASP. (2026). OWASP LLM Top 10. Open Web Application Security Project.
- Pretschner, A., et al. (2006). Distributed Usage Control. *Communications of the ACM*, 49(9), 39–44.
- Sheridan, T. B. (2002). *Humans and Automation: System Design and Research Issues*. Wiley.
- Stanford HAI. (2025). Artificial Intelligence Index Report 2025. Stanford University Human-Centered AI.
- Weill, P., and Ross, J. W. (2004). *IT Governance: How Top Performers Manage IT Decision Rights for Superior Results*. Harvard Business School Press.
- Xu, Y., et al. (2024). MultiAgentBench: Evaluating the Collaboration and Competition of LLM Agents. arXiv:2503.01935.